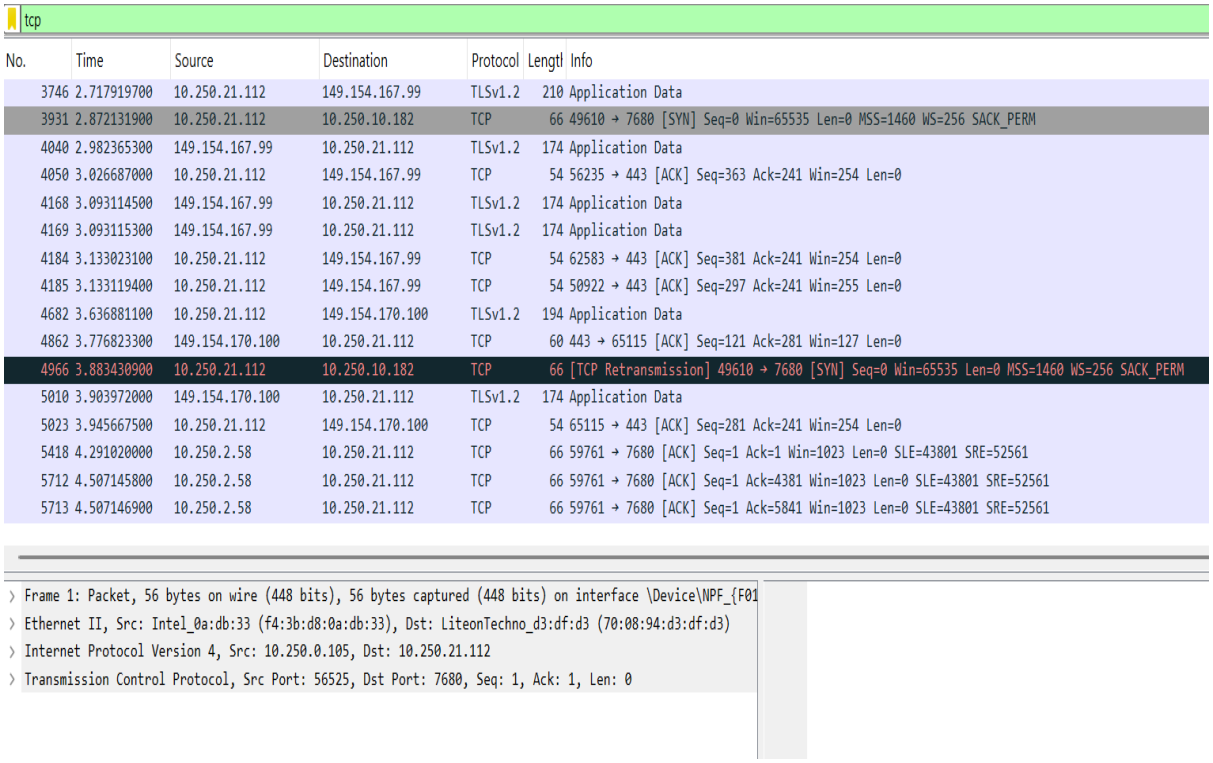
Week 1 (2520090044- S. Chakrika)

Task1

1.Apply the filter tcp and record the number of TCP packets.

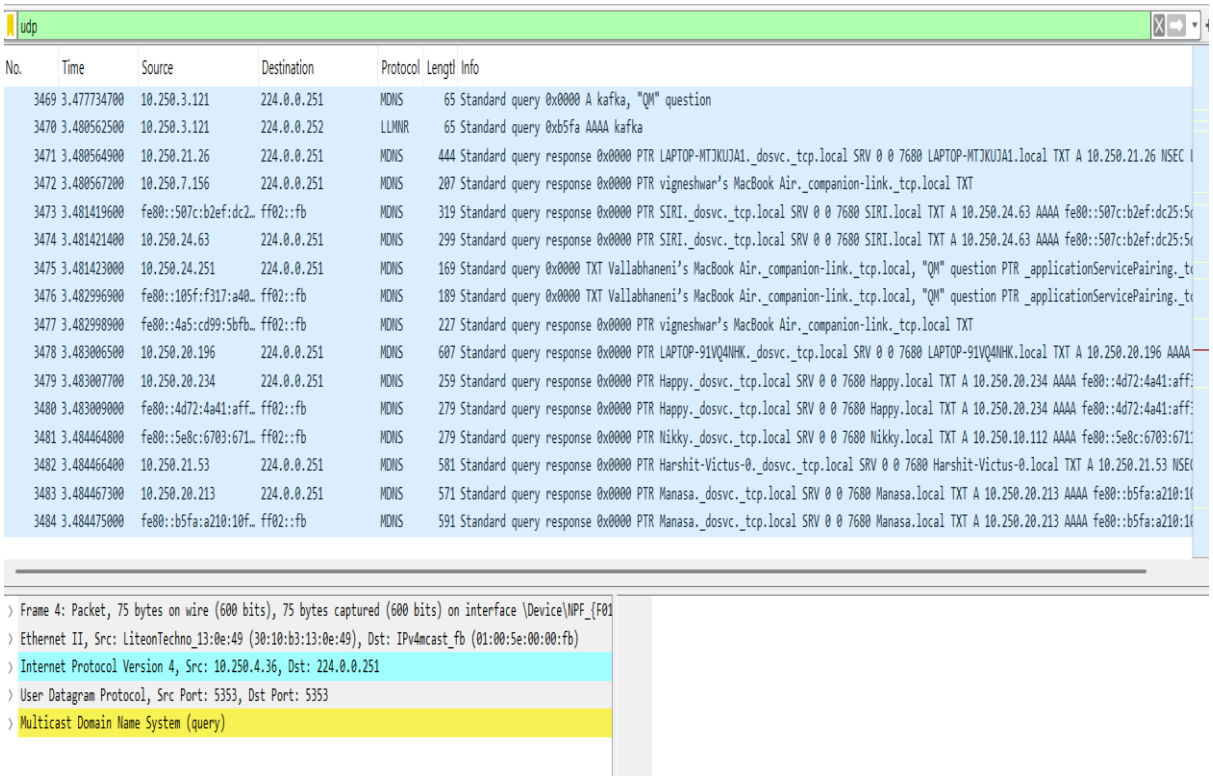


tcp

No.	Time	Source	Destination	Protocol	Lengtl	Info
3746	2.717919700	10.250.21.112	149.154.167.99	TLSv1.2	210	Application Data
3931	2.872131900	10.250.21.112	10.250.10.182	TCP	66	49610 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
4040	2.982365300	149.154.167.99	10.250.21.112	TLSv1.2	174	Application Data
4050	3.026687000	10.250.21.112	149.154.167.99	TCP	54	56235 → 443 [ACK] Seq=363 Ack=241 Win=254 Len=0
4168	3.093114500	149.154.167.99	10.250.21.112	TLSv1.2	174	Application Data
4169	3.093115300	149.154.167.99	10.250.21.112	TLSv1.2	174	Application Data
4184	3.133023100	10.250.21.112	149.154.167.99	TCP	54	62583 → 443 [ACK] Seq=381 Ack=241 Win=254 Len=0
4185	3.133119400	10.250.21.112	149.154.167.99	TCP	54	50922 → 443 [ACK] Seq=297 Ack=241 Win=255 Len=0
4682	3.636881100	10.250.21.112	149.154.170.100	TLSv1.2	194	Application Data
4862	3.776823300	149.154.170.100	10.250.21.112	TCP	60	443 → 65115 [ACK] Seq=121 Ack=281 Win=127 Len=0
4966	3.883430900	10.250.21.112	10.250.10.182	TCP	66	[TCP Retransmission] 49610 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
5010	3.903972000	149.154.170.100	10.250.21.112	TLSv1.2	174	Application Data
5023	3.945667500	10.250.21.112	149.154.170.100	TCP	54	65115 → 443 [ACK] Seq=281 Ack=241 Win=254 Len=0
5418	4.291020000	10.250.2.58	10.250.21.112	TCP	66	59761 → 7680 [ACK] Seq=1 Ack=1 Win=1023 Len=0 SLE=43801 SRE=52561
5712	4.507145800	10.250.2.58	10.250.21.112	TCP	66	59761 → 7680 [ACK] Seq=1 Ack=4381 Win=1023 Len=0 SLE=43801 SRE=52561
5713	4.507146900	10.250.2.58	10.250.21.112	TCP	66	59761 → 7680 [ACK] Seq=1 Ack=5841 Win=1023 Len=0 SLE=43801 SRE=52561


```
› Frame 1: Packet, 56 bytes on wire (448 bits), 56 bytes captured (448 bits) on interface \Device\NPF_{F01
› Ethernet II, Src: Intel_0a:db:33 (f4:3b:d8:0a:db:33), Dst: LiteonTechno_d3:df:d3 (70:08:94:d3:df:d3)
› Internet Protocol Version 4, Src: 10.250.0.105, Dst: 10.250.21.112
› Transmission Control Protocol, Src Port: 56525, Dst Port: 7680, Seq: 1, Ack: 1, Len: 0
```

2. Apply the filter udp and record the number of UDP packets



udp

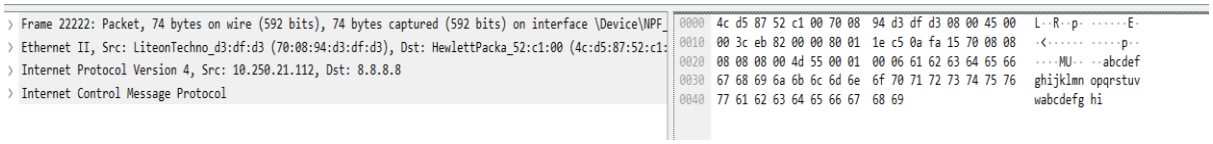
No.	Time	Source	Destination	Protocol	Lengtl	Info
3469	3.477734700	10.250.3.121	224.0.0.251	MDNS	65	Standard query 0x0000 A kafka, "QM" question
3470	3.480562500	10.250.3.121	224.0.0.252	LLMNR	65	Standard query 0xb5fa AAAA kafka
3471	3.480564900	10.250.21.26	224.0.0.251	MDNS	444	Standard query response 0x0000 PTR LAPTOP-MTJKUJA1._dosvc._tcp.local SRV 0 0 7680 LAPTOP-MTJKUJA1.local TXT A 10.250.21.26 NSEC
3472	3.480567200	10.250.7.156	224.0.0.251	MDNS	207	Standard query response 0x0000 PTR vigneshwar's MacBook Air._companion-link._tcp.local TXT
3473	3.481419600	fe80::507c:b2ef:dc2…	ff02::fb	MDNS	319	Standard query response 0x0000 PTR SIRI._dosvc._tcp.local SRV 0 0 7680 SIRI.local TXT A 10.250.24.63 AAAA fe80::507c:b2ef:dc25:5
3474	3.481421400	10.250.24.63	224.0.0.251	MDNS	299	Standard query response 0x0000 PTR SIRI._dosvc._tcp.local SRV 0 0 7680 SIRI.local TXT A 10.250.24.63 AAAA fe80::507c:b2ef:dc25:5
3475	3.481423000	10.250.24.251	224.0.0.251	MDNS	169	Standard query 0x0000 TXT Vallabhaneni's MacBook Air._companion-link._tcp.local, "QM" question PTR _applicationServicePairing._t
3476	3.482996900	fe80::105f:f317:a40…	ff02::fb	MDNS	189	Standard query 0x0000 TXT Vallabhaneni's MacBook Air._companion-link._tcp.local, "QM" question PTR _applicationServicePairing._t
3477	3.482998900	fe80::4a5:cd99:5bfb…	ff02::fb	MDNS	227	Standard query response 0x0000 PTR vigneshwar's MacBook Air._companion-link._tcp.local TXT
3478	3.483006500	10.250.20.196	224.0.0.251	MDNS	607	Standard query response 0x0000 PTR LAPTOP-91VQ4NHK._dosvc._tcp.local SRV 0 0 7680 LAPTOP-91VQ4NHK.local TXT A 10.250.20.196 AAAA
3479	3.483007700	10.250.20.234	224.0.0.251	MDNS	259	Standard query response 0x0000 PTR Happy._dosvc._tcp.local SRV 0 0 7680 Happy.local TXT A 10.250.20.234 AAAA fe80::4d72:4a41:aff:
3480	3.483009000	fe80::4d72:4a41:aff…	ff02::fb	MDNS	279	Standard query response 0x0000 PTR Happy._dosvc._tcp.local SRV 0 0 7680 Happy.local TXT A 10.250.20.234 AAAA fe80::4d72:4a41:aff:
3481	3.484464800	fe80::5e8c:6703:671…	ff02::fb	MDNS	279	Standard query response 0x0000 PTR Nikky._dosvc._tcp.local SRV 0 0 7680 Nikky.local TXT A 10.250.10.112 AAAA fe80::5e8c:6703:671
3482	3.484466400	10.250.21.53	224.0.0.251	MDNS	581	Standard query response 0x0000 PTR Harshit-Victus-0._dosvc._tcp.local SRV 0 0 7680 Harshit-Victus-0.local TXT A 10.250.21.53 NSE
3483	3.484467300	10.250.20.213	224.0.0.251	MDNS	571	Standard query response 0x0000 PTR Manasa._dosvc._tcp.local SRV 0 0 7680 Manasa.local TXT A 10.250.20.213 AAAA fe80::b5fa:a210:1
3484	3.484475000	fe80::b5fa:a210:10f…	ff02::fb	MDNS	591	Standard query response 0x0000 PTR Manasa._dosvc._tcp.local SRV 0 0 7680 Manasa.local TXT A 10.250.20.213 AAAA fe80::b5fa:a210:1


```
› Frame 4: Packet, 75 bytes on wire (600 bits), 75 bytes captured (600 bits) on interface \Device\NPF_{F01
› Ethernet II, Src: LiteonTechno_13:0e:49 (30:10:b3:13:0e:49), Dst: IPv4mcast_fb (01:00:5e:00:00:fb)
› Internet Protocol Version 4, Src: 10.250.4.36, Dst: 224.0.0.251
› User Datagram Protocol, Src Port: 5353, Dst Port: 5353
› Multicast Domain Name System (query)
```

3. Apply the filter icmp and record the number of ICMP packets

icmp

No.	Time	Source	Destination	Protocol	Length	Info
22222	14.397489300	10.250.21.112	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=6/1536, ttl=128 (reply in 22226)
22226	14.421687500	8.8.8.8	10.250.21.112	ICMP	74	Echo (ping) reply id=0x0001, seq=6/1536, ttl=115 (request in 22222)
23698	15.418804800	10.250.21.112	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=7/1792, ttl=128 (reply in 23801)
23801	15.459176000	8.8.8.8	10.250.21.112	ICMP	74	Echo (ping) reply id=0x0001, seq=7/1792, ttl=115 (request in 23698)
25423	16.432054100	10.250.21.112	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=8/2048, ttl=128 (reply in 25459)
25459	16.462857900	8.8.8.8	10.250.21.112	ICMP	74	Echo (ping) reply id=0x0001, seq=8/2048, ttl=115 (request in 25423)
27013	17.442553000	10.250.21.112	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=9/2304, ttl=128 (reply in 27075)
27075	17.497133400	8.8.8.8	10.250.21.112	ICMP	74	Echo (ping) reply id=0x0001, seq=9/2304, ttl=115 (request in 27013)



4. Identify the top five protocols using Statistics → Protocol Hierarchy.

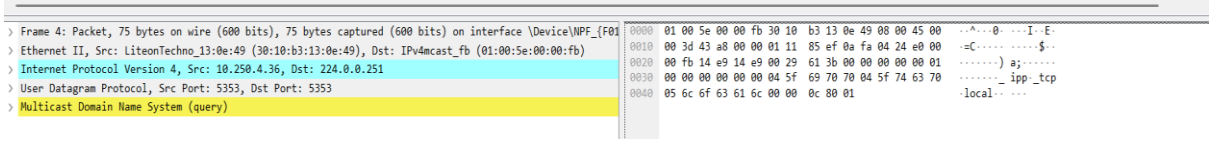
Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	2143	100.0	503715	1156 k	0	0	0	2143
Ethernet	100.0	2143	6.0	30002	68 k	0	0	0	2143
Internet Protocol Version 6	38.5	826	6.6	33040	75 k	0	0	0	826
User Datagram Protocol	38.5	826	1.3	6608	15 k	0	0	0	826
Simple Service Discovery Protocol	0.2	4	0.1	454	1042	4	454	1042	4
Multicast Domain Name System	37.1	795	32.2	162171	372 k	795	162171	372 k	795
Link-local Multicast Name Resolution	0.9	19	0.1	617	1416	19	617	1416	19
eXtensible Markup Language	0.2	4	0.5	2560	5877	4	2560	5877	4
DHCPv6	0.2	4	0.1	313	718	4	313	718	4
Internet Protocol Version 4	61.5	1317	5.2	26340	60 k	0	0	0	1317
User Datagram Protocol	61.5	1317	2.1	10536	24 k	0	0	0	1317
Simple Service Discovery Protocol	2.4	51	1.5	7349	16 k	51	7349	16 k	51
NetBIOS Name Service	1.6	34	0.4	1880	4316	34	1880	4316	34
NetBIOS Datagram Service	0.2	5	0.1	410	941	0	0	0	5
SMB (Server Message Block Protocol)	0.2	5	0.1	487	1118	0	0	0	5
SMB MailSlot Protocol	0.2	5	0.0	125	286	0	0	0	5
Microsoft Windows Browser Protocol	0.2	5	0.0	57	130	5	57	130	5
Multicast Domain Name System	48.7	1044	41.3	207842	477 k	1044	207842	477 k	1044
Link-local Multicast Name Resolution	1.8	39	0.2	1067	2449	39	1067	2449	39
eXtensible Markup Language	0.2	4	0.5	2560	5877	4	2560	5877	4
Dynamic Host Configuration Protocol	0.3	6	0.4	1892	4343	6	1892	4343	6
Data	6.3	134	3.3	16467	37 k	134	16467	37 k	134

Task 2: Ethernet frame inspection

6. Apply the filter eth.

eth

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	Intel_ec:b1:9d	Broadcast	ARP	56	Who has 10.250.20.186? Tell 10.250.6.108
2	0.000001000	AzureWaveTec_ef:5b:…	Broadcast	ARP	42	Who has 10.250.2.225? Tell 10.250.0.201
3	0.000001400	ChongqingFug_db:35:…	Broadcast	ARP	56	Who has 10.250.14.235? Tell 10.250.27.22
4	0.000002100	10.250.4.36	224.0.0.251	MDNS	75	Standard query 0x0000 PTR _ipp._tcp.local, "QU" question
5	0.000002900	fe80::d19a:3533:c35…	ff02::fb	MDNS	108	Standard query 0x0000 A ADITYA.local, "QM" question A 10.250.0.164
6	0.000003600	fe80::d19a:3533:c35…	ff02::fb	MDNS	120	Standard query 0x0000 AAAA ADITYA.local, "QM" question AAAA fe80::f8c2:e398:155c:98b8
7	0.000004000	10.250.5.35	224.0.0.251	MDNS	100	Standard query 0x0000 AAAA ADITYA.local, "QM" question AAAA fe80::f8c2:e398:155c:98b8
8	0.000004300	fe80::f315:947:62b2…	ff02::fb	MDNS	95	Standard query 0x0000 PTR _ipp._tcp.local, "QU" question
9	0.000004700	AzureWaveTec_72:b9:…	Broadcast	ARP	56	Who has 10.250.1.1? Tell 10.250.9.191
10	0.000006100	10.250.1.208	224.0.0.251	MDNS	310	Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0 0 60585 Reddy.local TXT A 10.250.1.
11	0.000006700	10.250.28.48	224.0.0.251	MDNS	225	Standard query response 0x0000 PTR, cache flush iPhone-3.local PTR, cache flush iPhone-3.local NSEC, cache flush 3.5.E.8.3.7.B.2.
12	0.000007200	10.250.4.36	224.0.0.251	MDNS	77	Standard query 0x0000 PTR _uscan._tcp.local, "QU" question
13	0.005501400	fe80::f315:947:62b2…	ff02::fb	MDNS	97	Standard query 0x0000 PTR _uscan._tcp.local, "QU" question
14	0.005503800	fe80::d9:f654:2b73:…	ff02::fb	MDNS	245	Standard query response 0x0000 PTR, cache flush iPhone-3.local PTR, cache flush iPhone-3.local NSEC, cache flush 3.5.E.8.3.7.B.2.
15	0.005507100	10.250.4.36	224.0.0.251	MDNS	78	Standard query 0x0000 PTR _uscans._tcp.local, "QU" question
16	0.005508200	10.250.20.106	224.0.0.251	MDNS	424	Standard query response 0x0000 TXT PTR Arjun's MacBook Air._companion-link._tcp.local TXT, cache flush SRV, cache flush 0 0 4953
17	0.005509900	fe80::cfe:9eed:56e4…	ff02::fb	MDNS	444	Standard query response 0x0000 TXT PTR Arjun's MacBook Air. companion-link. tcp.local TXT, cache flush SRV, cache flush 0 0 4953



7. Select five different packets and record their:

- Frame number

- Frame length

- Source MAC address

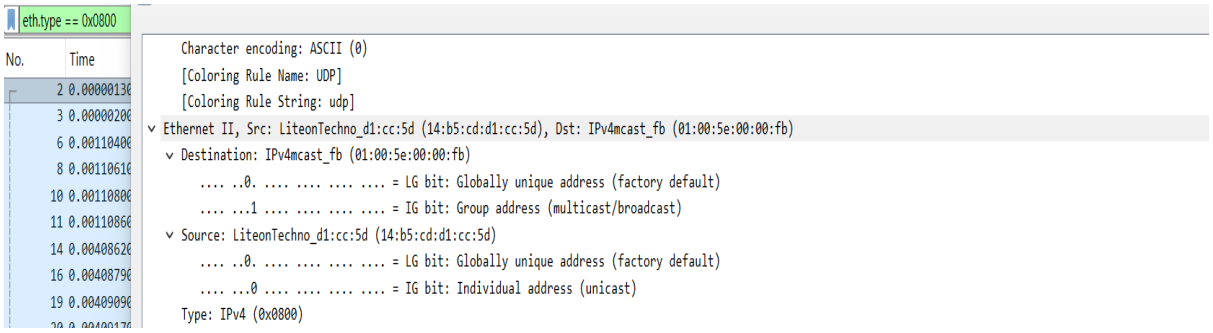
- Destination MAC address

- EtherType

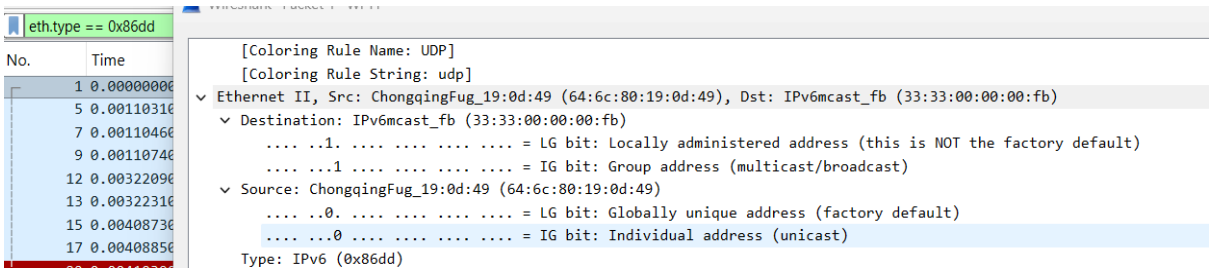
Frame No.	Frame Length	Source MAC Address	Destination MAC Address	EtherType
1	56 bytes	9c:67:d6:ec:b1:9d	ff:ff:ff:ff:ff:ff	ARP
2	42 bytes	f8:3d:c6:ef:5b:ae	ff:ff:ff:ff:ff:ff	ARP
3	56 bytes	b4:b5:b6:db:35:9f	ff:ff:ff:ff:ff:ff	ARP
4	75 bytes	30:10:b3:13:0e:49	01:00:5e:00:00:fb	IPv4
5	108 bytes	b8:1e:a4:9d:6b:ad	33:33:00:00:00:fb	IPv6

8. Identify packets carrying IPv4 and IPv6 traffic

IPV4:

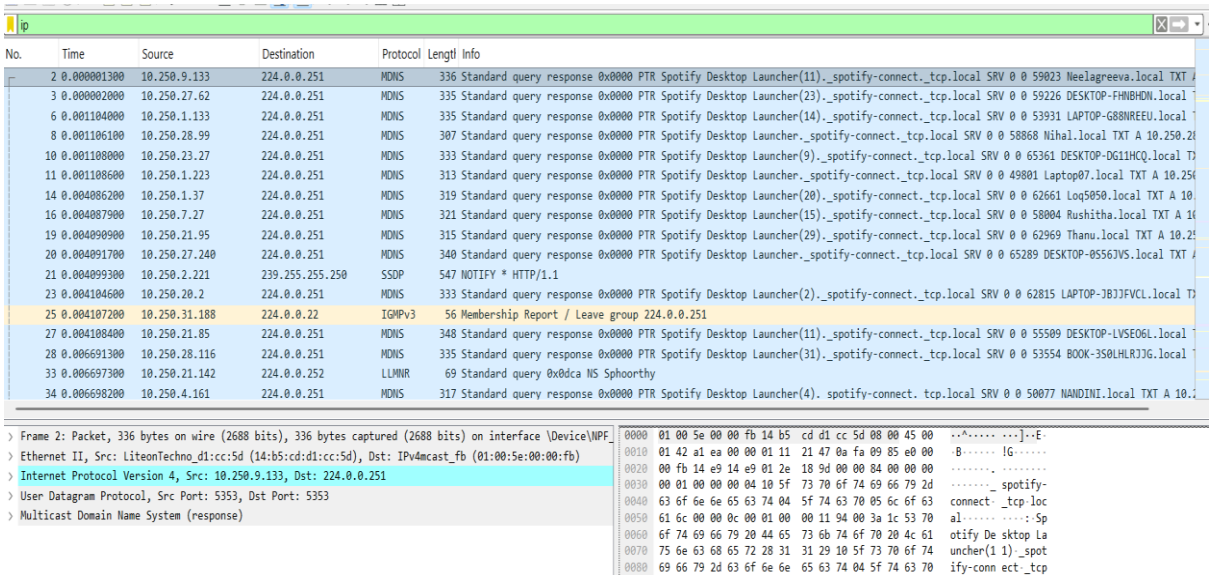


IPV6:



Task 3: IPv4 packet inspection

9. Apply the filter ip



10. Identify five IPv4 packets and record

- Source IP address
- Destination IP address
- Header length
- Total length
- TTL
- Identification field

Frame No.	Source IP Address	Destination IP Address	Header Length	Total Length	TTL	Identification
2	10.250.9.133	224.0.0.251	20 bytes	322 bytes	1	41450
3	10.250.27.62	224.0.0.251	20 bytes	321 bytes	1	62183
6	10.250.1.133	224.0.0.251	20 bytes	321 bytes	1	6083
8	10.250.28.99	224.0.0.251	20 bytes	293 bytes	1	5392
10	10.250.23.27	224.0.0.251	20 bytes	319 bytes	1	872

11. Find the packet with the maximum IPv4 total length.

From the Selected 5 IPv4

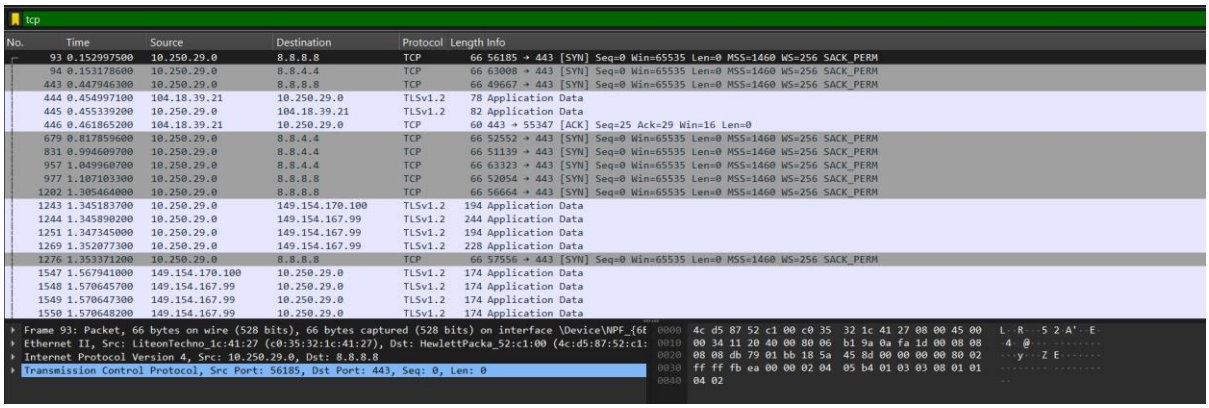
- Frame Number : 2
- Source IP: 10.250.9.133
- Destination IP: 224.0.0.251

12. Identify packets with different TTL values.

All five selected packets have TTL = 1.

Task 4: TCP header inspection

13. Apply TCP filter



14. Select five TCP packets and record:

- Source port

- Destination port

- Sequence number

- Acknowledgement number

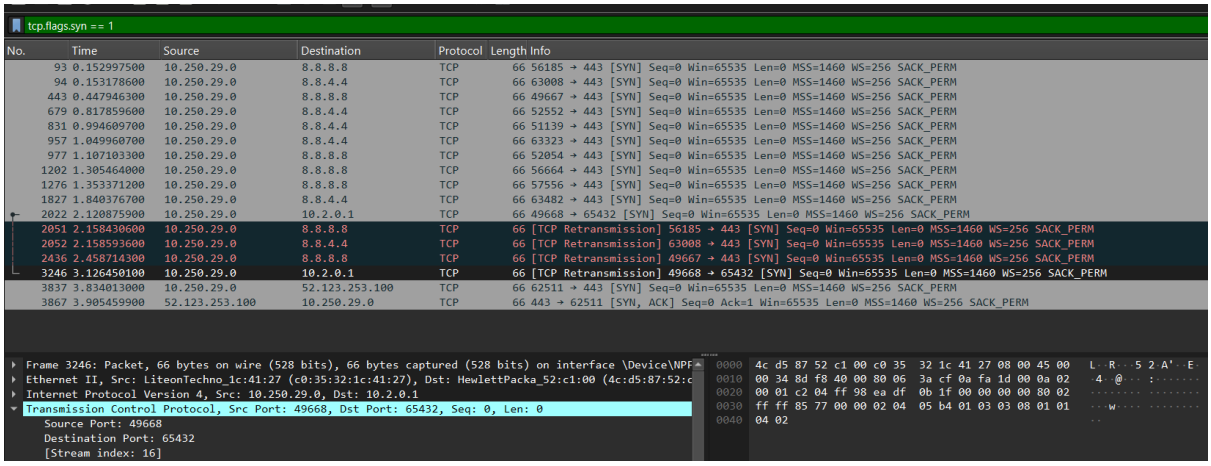
- TCP header length

- Payload length

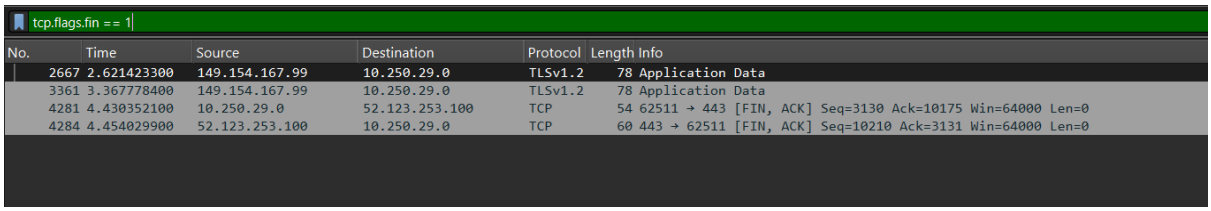
S.No.	Frame	Source	Destination	Sequence	Acknowledgement	TCP	Payload

	No.	Port	Port	Number	Number	Header Length	Length
1	94	63008	443	0	0	32 bytes	0 bytes
2	443	49667	443	0	0	32 bytes	0 bytes
3	679	52552	443	0	0	32 bytes	0 bytes
4	1276	57556	443	0	0	32 bytes	0 bytes
5	3576	52555	443	315	241	20 bytes	0 bytes

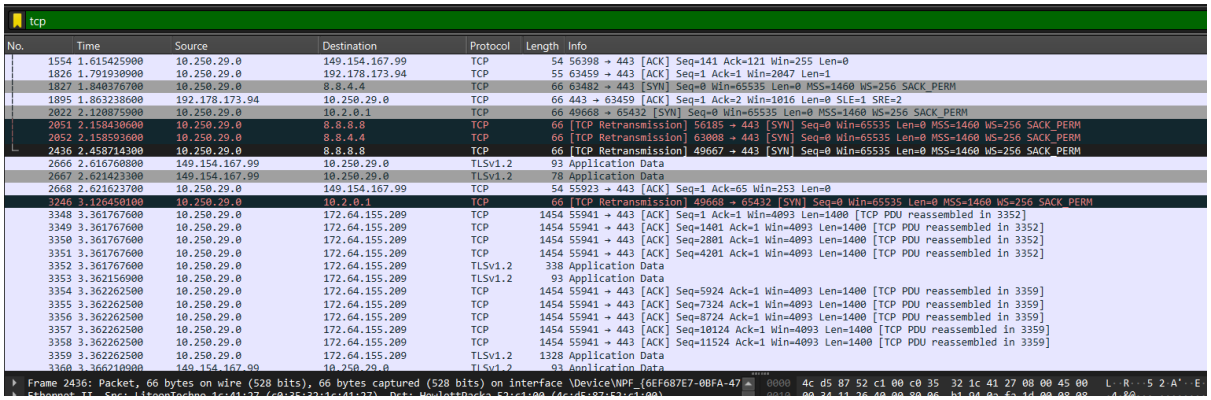
15. Apply the filter tcp.flags.syn == 1 and identify TCP connection establishment packets.



16. Apply the filter tcp.flags.fin == 1 and identify connection termination packets



17. Identify packets containing TCP options.



S.No.	Frame Number	TCP Header Length	TCP Options Identified
1	2051	32 bytes	MSS, NOP, Window Scale, NOP, NOP
2	3246	32 bytes	MSS, NOP, Window Scale, NOP, NOP
3	2436	32 bytes	MSS, NOP, Window Scale, NOP, NOP

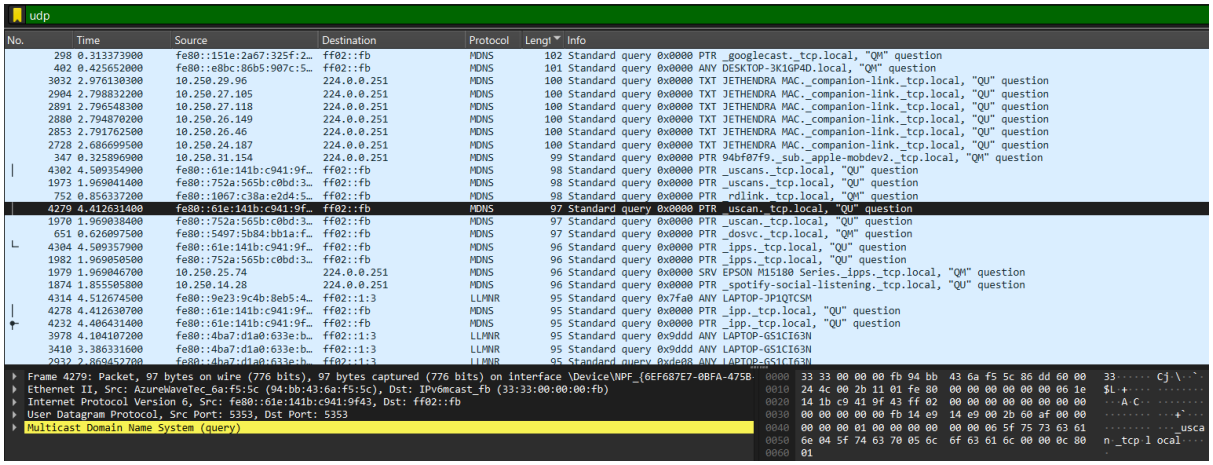
18. Find the TCP packet with the maximum payload length.

S.No.	Frame Number	TCP Segment Length
1	4282	35 bytes
2	445	28 bytes
3	2667	24 bytes
4	4074	31 bytes
5	4280	825 bytes

The Max payload length is 825 bytes of Frame 4280.

Task 5: UDP header inspection

19. Apply the filter udp.



20. Select five UDP packets and record:

- Source port

- Destination port

- UDP length

- Payload length

S.No.	Frame No.	Source Port	Destination Port	UDP Length	Payload Length
1	4279	5353	5353	43 bytes	35 bytes
2	2904	5353	5353	66 bytes	58 bytes
3	347	5353	5353	65 bytes	57 bytes
4	4278	5353	5353	41 bytes	33 bytes
5	2853	5353	5353	66 bytes	58 bytes

21. Identify the UDP packet with the maximum payload length.

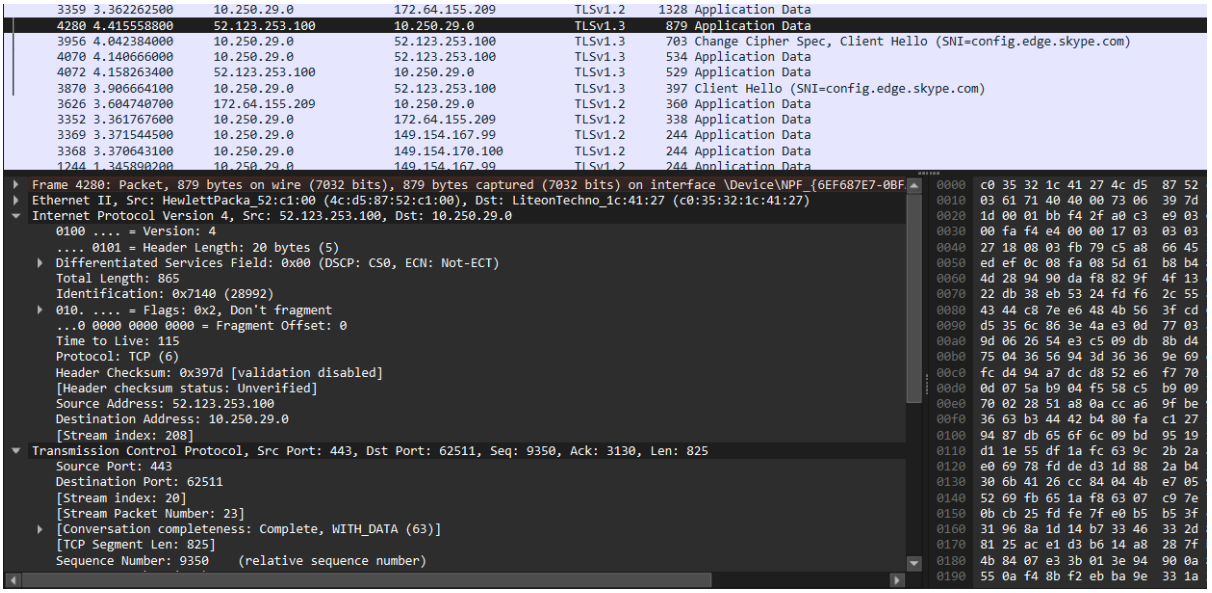
From the selected 5 Frames, Frame Number 2904 has Max Payload Length of 58 bytes.

22. Find the UDP packets carrying DNS traffic using the filter



Task 6: Encapsulation and overhead calculation

23. Select one TCP packet containing application data. Record the Ethernet, IPv4 and TCP header lengths.



Protocol	Header Length
Ethernet II	14 bytes
IPv4	20 bytes
TCP	20 bytes

24. Calculate the total protocol header overhead for the selected packet.

Formula:

$$Total\ Overhead = Ethernet + IPv4 + TCP$$

$$= 14 + 20 + 20 = 54 \text{ bytes.}$$

25. Calculate the application payload size.

The TCP Segment Len for Frame 4280 is **825 bytes.**

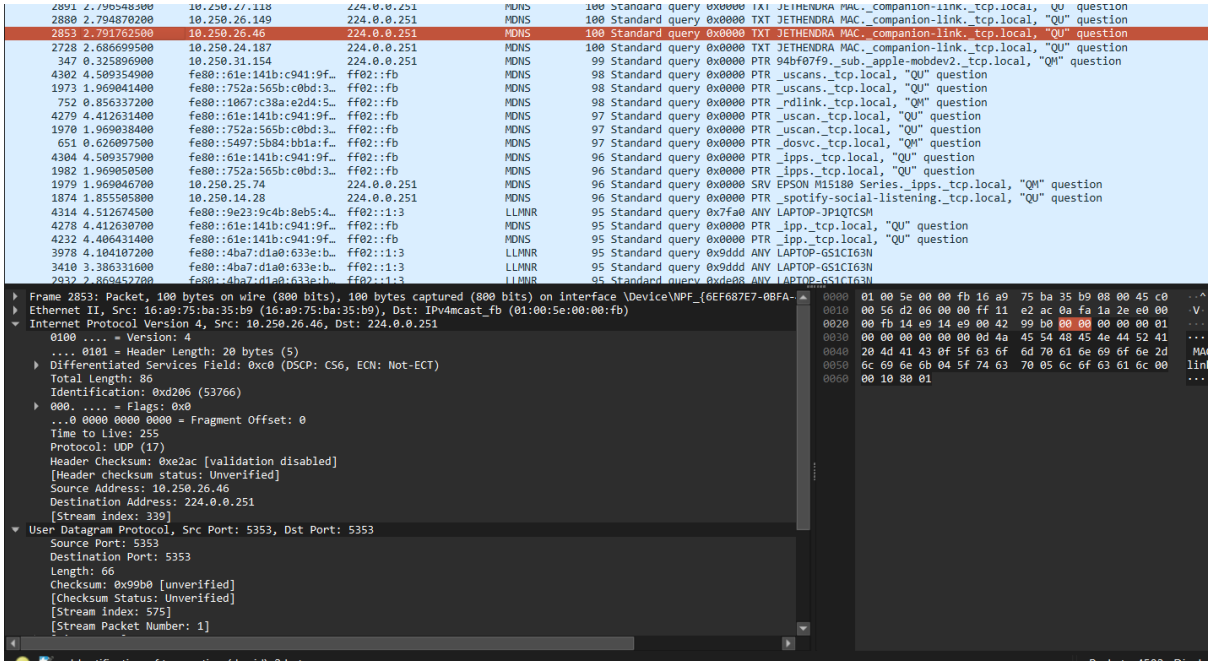
26. Calculate the payload efficiency percentage.

Formula:

*Efficiency = (Payload/Total Frame Length) * 100*

= (825/879) * 100 = 93.86%

27. Repeat the same calculations for one UDP packet.



Total Overhead = 14+20+8 = 42 bytes

Payload = 66-8 = 58

Efficiency = (58/100) * 100 = 58%

Final UDP Observation Table

Parameter	Value
Frame Number	2853
Frame Length	100 bytes
Ethernet Header	14 bytes
IPv4 Header	20 bytes
UDP Header	8 bytes
Total Overhead	42 bytes
Application Payload	58 bytes
Payload Efficiency	58%

28. Compare the calculated overhead and payload efficiency of the selected TCP and UDP packets.

TCP: Frame number - 4280

UDP: Frame number - 2853

Parameter	TCP (Frame 4280)	UDP (Frame 2853)
Frame Length	879 bytes	100 bytes
Ethernet Header	14 bytes	14 bytes
IPv4 Header	20 bytes	20 bytes
Transport Header	20 bytes	8 bytes
Total Header Overhead	54 bytes	42 bytes
Application Payload	825 bytes	58 bytes
Payload Efficiency	93.86%	58%